

Internal Research Report: Do Temporal Network Signals Predict How Scientific Concepts Spread Across Disciplines?

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Contents

1	Overview and goals	3
2	Iteration 1: Wide screen of five rival mechanisms	3
2.1	Strategy and rationale	3
2.2	Experiment 1: Does citing a concept “as your own” predict its spread?	3
2.3	Experiment 3: Do diverse topic ties predict concept spread?	5
2.4	Experiment 4: Where a concept lands early versus how broadly it spreads	5
2.5	Failed artifacts	5
2.6	Dead ends from iteration 1	5
2.7	Review and hypothesis update	6
3	Iteration 2: Gateway retention scaled up, and field entry	6
3.1	Strategy and rationale	6
3.2	Experiment 5: Panel construction and gateway retention test	6
3.3	Experiment 6: Where new scientific concepts spread next	7
3.4	Evaluation 1: Stress-testing the gateway-field retention lead	8
3.5	Dataset 2: When research concepts were officially recognised	8
3.6	Research 1: How our results compare with related papers	8
3.7	Dead ends from iteration 2	8
3.8	Review and hypothesis update	8
4	Iteration 3: The retained frontier and the indicator screen	9
4.1	Strategy and rationale	9
4.2	Experiment 7: Do concepts spread from fields that keep them?	9
4.3	Experiment 8: Which early network signals travel across fields	9
4.4	Evaluation 2: Auditing the record before the paper	11
4.5	Research 2: Prior art and venue check	11
4.6	Failed artifact: Experiment 9	13
4.7	Dead ends from iteration 3	13
4.8	Review and hypothesis update	13

5	Iteration 4: Fresh-cohort confirmation and RQ2 trajectories	13
5.1	Strategy and rationale	13
5.2	Experiment 10: Fresh-cohort confirmation of OPEN	13
5.3	Experiment 11: Within-concept closure test (incomplete)	15
5.4	Experiment 12: Breadth decomposition and trajectories	15
5.5	Evaluation 3: Record fixes and openness robustness	15
5.6	Research 3: Novelty check	17
5.7	Dead ends from iteration 4	17
5.8	Review and hypothesis update	17
6	Iteration 5: Vocabulary-free confirmation and confound analysis	17
6.1	Strategy and rationale	17
6.2	Experiment 13: Does the churn signal hold for brand-new phrases?	17
6.3	Experiment 14: Cheng’s consistency—a size effect, not a reach predictor	18
6.4	Experiment 15: Why churning concepts spread; Experiment 11 completion	20
6.5	Evaluation 4: Record repair and openness evidence pool	21
6.6	Experiment 16: Is neighbourhood churn real or thin-sample noise?	21
6.7	Dead ends from iteration 5	21
6.8	Review and hypothesis update	23
7	What the run learned and what is still open	23
7.1	Confirmed findings	23
7.2	Dead ends and negative results (cumulative)	24
7.3	What is still open	24
7.4	Run bookkeeping	25

1 Overview and goals

This report is the chronological record of a five-iteration research run investigating whether early structural signals in a scientific concept’s topic co-occurrence network anticipate later cross-field breadth, beyond simple measures of volume and reach. Using the OpenAlex corpus (476 million works, 2026-09-23 snapshot), the run identified 12,499 concepts, tracked 27,393 adoption episodes across 26 fields, and screened 53 early network indicators against a five-feature popularity baseline.

The run pursued two research questions:

- **RQ1:** Which early temporal network indicators anticipate cross-disciplinary breadth, validated on held-out field groups?
- **RQ2:** Do concepts spread next to fields related to the ones currently retaining them, extending the principle of relatedness from economic complexity?

The primary outcome is rarefied field breadth (O_{2r} , $m = 50$): the expected number of distinct venue fields among a fixed-size random draw of papers from a concept’s publications in years $t_0 + 6$ to $t_0 + 8$.

2 Iteration 1: Wide screen of five rival mechanisms

2.1 Strategy and rationale

Iteration 1 was a wide screen. The preceding hypothesis-stage review had computed a reliability of approximately 0.32 for the naturalisation gap A_h^* from the probe, and only 1 of 8 confidence intervals excluded zero. The strategy therefore raced five rival mechanisms on a single frozen dev panel: lineage naturalisation (A_h^*), social reach through unconnected author groups, structural diversity of new co-occurrence neighbours, frequency-free selectivity, and landing in gateway fields. Five artifacts were commissioned; three completed.

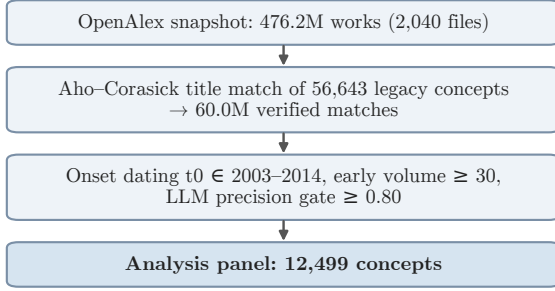
2.2 Experiment 1: Does citing a concept “as your own” predict its spread?

Experiment 1 (`gen_art_experiment_1`, iteration 1) screened the background-adjusted naturalisation gap A_h^* on the frozen 78-concept dev panel (48 concepts after dropping those with t_0 outside 2003–2009 and those with a sealed home field). The shared OpenAlex credit pool ran dry (139 own credits spent), so yearly counts came from OpenAlex exactly, but field labels, concept papers, citation lineage and background-reference fields came from free Semantic Scholar data.

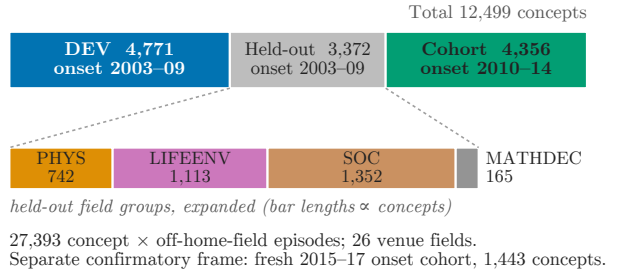
Result. A_h^* did *not* survive the pre-registered rule:

Other findings. A_h^* ’s within-field Spearman with O_{2r} flips sign across groups: Medicine +0.45, CS −0.18; median A_h^* computed from `features.csv` is negative in every group (BGM −0.26, CS −0.30, Eng −0.04, Med −0.18). R^2 of raw lineage log-odds ratio on the background log-odds ratio is 0.66; the background log-odds ratio is positive for 48/48 concepts, so raw lineage is mostly homophily. REML $\tau_c = 0.29$, $\tau_{cj} = 0.65$, confirming that between-concept variance in cross-field behaviour is mostly at the concept $\times$ field level. None of the 14 candidate and foil features beats B5.

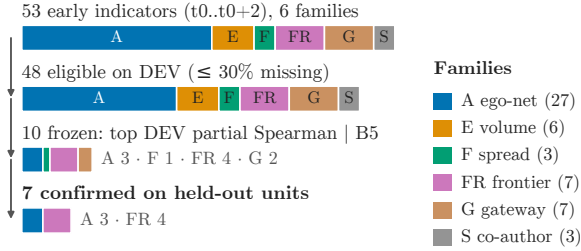
(a) Concept identification



(b) Panel structure



(c) Indicator screen (breadth outcome)



(d) Research questions

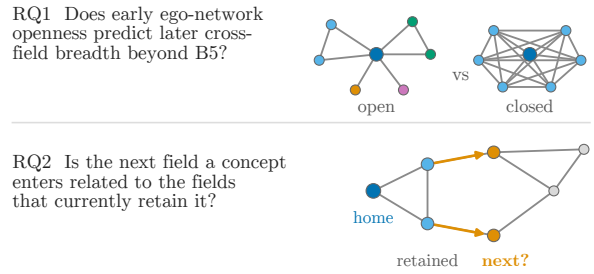


Figure 1: Overview of the study design. (a) Concept identification: 56,643 legacy OpenAlex concepts are matched by title (Aho–Corasick) against a 476.2M-work OpenAlex snapshot, giving 60.0M verified matches; concepts are onset-dated ($t_0 \in 2003-2014$), required to have early volume ≥ 30 and to pass an LLM precision gate ≥ 0.80 , yielding an analysis panel of 12,499 concepts. (b) Panel structure, bar lengths proportional to concept counts: DEV (blue; CS, Eng, BGM and Med homes, onset 2003–09) 4,771, held-out field groups (grey; onset 2003–09) 3,372, and the 2010–14 onset cohort (green) 4,356. The expanded lower bar splits the held-out block into PHYS 742, LIFEENV 1,113, SOC 1,352 and MATHDEC 165. The panel contains 27,393 concept $\times$ off-home-field episodes over 26 venue fields; a separate fresh 2015–17 onset cohort of 1,443 concepts serves as a confirmatory frame. (c) Indicator screen for the breadth outcome (O2r_m50). Bars are coloured by indicator family with lengths proportional to indicator counts: 53 early indicators computed over $t_0..t_0+2$; 48 eligible on DEV; the 10 frozen on DEV; and the 7 confirmed on held-out units. (d) The two research questions as schematic illustrations.

Table 1: A_h^* screen result (Experiment 1, iteration 1). Source: `screen_result.json`.

Criterion	Value	Pass?
LOGO $\Delta\rho$ for O_{2r} over B5	−0.006 [−0.034, 0.017]	Fail
ρ_{B5}	0.834	—
Positive left-out groups	0 of 4	Fail
Split-half reliability (Spearman–Brown)	0.58	Fail (bar 0.6)
$ \rho $ with log early vol / early growth	0.14 / 0.18	Pass

2.3 Experiment 3: Do diverse topic ties predict concept spread?

Experiment 3 (`gen_art_experiment_3`, iteration 1) screened two co-occurrence emergence indicators (D_{ratio} and F_{res}) on the frozen dev panel (47 concepts). The backbone used full-corpus topic PMI per time slice (2000–04, 05–09, 10–14), with Leiden $\gamma = 3$ yielding 25/26/23 communities. D_z failed the size diagnostic (ρ with log volume -0.63), so the pre-declared fallback D_{ratio} was the primary.

Table 2: Experiment 3 screen results (iteration 1). Source: `screen_result.json`.

Indicator	$\Delta\rho$	90% CI	Groups +	SB	Pass?
D_{ratio}	+0.006	[−0.092, 0.135]	3/4	0.83	No
F_{res}	−0.060	[−0.158, 0.014]	1/4	0.44	No

B5 alone reached $\rho = 0.770$ with O_{2r} . No candidate survived the pre-registered rule; D_{ratio} was carried forward as the best available result. Exploratory partial Spearman of D_{ratio} given B5 was +0.335 [0.02, 0.65], permutation $p = 0.037$ —marginal, uncorrected, 1 of 12 tests.

2.4 Experiment 4: Where a concept lands early versus how broadly it spreads

Experiment 4 (`gen_art_experiment_4`, iteration 1) screened the gateway-landing indicator G on the dev panel (46 concepts, 34 with outcome windows). This artifact also produced the authoritative outcome tables for the iteration.

Table 3: Experiment 4 screen results (iteration 1). Source: `screen_result.json`.

Outcome	$\Delta\rho$ or ΔAUC	90% CI	Groups +	Pass?
O_{2r} (breadth)	+0.033	[−0.095, 0.168]	2/4	No
O_{2r} residualised on log N	+0.15	[0.000, 0.321]	4/4	—
O_1 (uptake)	$\Delta\text{AUC} +0.072$	[0.00, 0.16]	—	—

B5 reached $\rho = 0.327$ on this panel (CS 0.10, Eng 0.86, BGM 0.65, Med 0.57), far lower than in Experiments 1 and 3 due to top-200-source truncation. G did not survive the pre-registered rule.

Field-level gateway retention. The adopting field’s eigenvector centrality on the topic-relatedness backbone added $\Delta\text{AUC} = +0.103$ (95% CI [0.03, 0.17]) for retention on the 80-episode panel. This was the iteration’s only live lead.

2.5 Failed artifacts

Two artifacts failed (“REPL turn stalled”): `gen_art_dataset_1` (sealed held-out test set) and `gen_art_experiment_2` (candidate S : unconnected co-author groups). As a result, no held-out evaluation existed after iteration 1, and candidate S was untested.

2.6 Dead ends from iteration 1

- A_h^* naturalisation gap: fails every clause of the pre-registered rule.
- D_{ratio} and F_{res} : no candidate survives the rule.

- G gateway landing on O_{2r} : fails ($\Delta\rho = +0.033$).
- Background homophily explains 66–72% of raw lineage variance (measurement finding, not a dead end per se).
- CS-only co-occurrence growth indicators: positive in CS, negative or null in other fields.

2.7 Review and hypothesis update

The reviewer scored iteration 1 at 3 (blocking). Main objections: a conclusion contradicted the evidence (the “baseline ceiling” argument did not apply to Experiment 4, where $\rho_{B5} = 0.327$); A_h^* medians were misread; two failed artifacts were missing. The hypothesis moved to **deepen** the only live lead: field-level gateway retention ($\Delta\text{AUC} +0.10$, dev only, $n = 80$). Evidence state: *lead*. Confidence: *decreased*.

3 Iteration 2: Gateway retention scaled up, and field entry

3.1 Strategy and rationale

Iteration 2 moved the budget from more metrics to more samples, building a 12,499-concept / 27,393-episode zero-credit OpenAlex snapshot panel with a hash-sealed held-out split. The strategy tested two claims: (H1) the adopting field’s gateway centrality predicts concept retention beyond B5, field size, relatedness to home, density, and the field’s own retention propensity; (H2) a concept spreads next to fields related to the ones currently retaining it; plus (H3) concept-level gateway landing predicts volume-residualised breadth.

3.2 Experiment 5: Panel construction and gateway retention test

Experiment 5 (`gen_art_experiment_5`, iteration 2) built the authoritative S1 tables for the run: 12,499 concepts (DEV 4,771, held-out PHYS 742, LIFEENV 1,113, SOC 1,352, MATHDEC 165, cohort 4,356) and 27,393 episodes. Concepts were identified by Aho–Corasick title matching of 56,643 legacy OpenAlex concepts (levels 2–5) plus Wikidata aliases, with stemmed verification and a per-concept LLM precision gate (≥ 0.80). The spec was frozen on DEV and unsealed once.

Table 4: Panel composition (Experiment 5, iteration 2). Source: `frame_concepts.csv`.

Split	Concepts	Episodes
DEV (CS/Eng/BGM/Med, onset 2003–2009)	4,771	9,079
Cohort (onset 2010–2014, all fields)	4,356	9,799
Held-out Physical Sciences	742	1,662
Held-out Life & Environment	1,113	3,099
Held-out Social Sciences	1,352	3,320
Held-out Math & Decision Sci.	165	434
Total	12,499	27,393

H1 result: disconfirmed. On the full panel, gateway centrality adds $\Delta\text{AUC} -0.00001$ (95% CI $[-0.0006, +0.0003]$); DerSimonian–Laird pooled -0.00004 ($I^2 = 0$). The baseline ladder shows that the signal vanishes once the field’s leave-concept-out retention propensity is controlled ($+0.0019$ on DEV drops to $+0.00003$; negative on all held-out groups).

H3 result: small and fragile. Held-out partial ρ : $G = 0.030$, $G_A = 0.026$, $G_{\text{btw}} = 0.046$ (Holm $p = 0.0045$); DL pooled $G = 0.068$ [0.029, 0.107]. But the concept-bootstrap 95% CI of pooled G is $[-0.006, 0.065]$, which includes zero. DEV-to-held-out shrinkage is about $4\times$ (DEV $G = 0.138$).

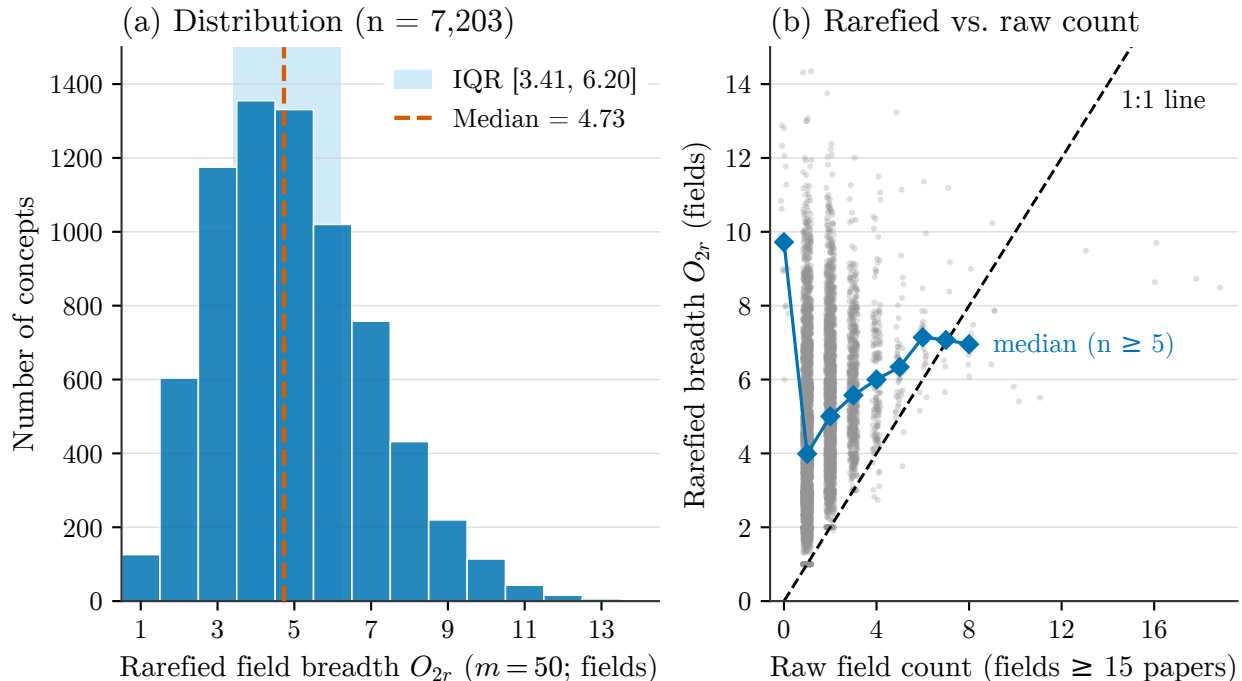


Figure 2: Distribution of the primary outcome, rarefied field breadth (O_{2r} , $m = 50$), over the 7,203 of 12,499 frame concepts that have at least 50 papers in the $t_0+6..t_0+8$ outcome window. (a) Histogram of O_{2r} in 1-field bins. The dashed orange line marks the median (4.73) and the light-blue band the interquartile range [3.41, 6.20]. (b) O_{2r} against the raw field count; rarefied breadth is nearly independent of outcome-window volume ($\rho = -0.06$).

3.3 Experiment 6: Where new scientific concepts spread next

Experiment 6 (`gen.art.experiment_6`, iteration 2) tested field entry on 653 newborn concepts (274 DEV, 369 held-out) using the frozen 26-field PMI backbone.

H2 entry result. Held-out LR = 71.7 ($p = 2 \times 10^{-17}$), $d_0 = 0.302$ [0.240, 0.369], positive in Physical, LifeEnv, Social and Cohort, DL pooled 0.284 [0.216, 0.352] ($I^2 = 0$). Label-permutation $p = 0.001$, rewired-backbone $p = 0.015$. However, the gateway *weighting* added nothing beyond plain retaining relatedness (M3 vs M1 permutation $p = 0.17$ held-out).

Trajectories. DTW k -medoids $k = 2$ gave stable classes (bootstrap ARI 1.0) but the HMM gave ARI = 0.094 against DTW classes—the two-class typology did not reproduce.

Ordering. 57 of 87 non-tied cases (65.5%) had the first retained gateway field preceding the calibrated entropy take-off, but the lead-lag forward regression was negative ($b = -0.028$, $p = 0.0007$ held-out), with a significant pre-trend ($\text{ev-3} = -0.072$, $p = 0.0002$). The ordering result was later reclassified as **mixed**.

3.4 Evaluation 1: Stress-testing the gateway-field retention lead

Evaluation 1 (`gen_art_evaluation_1`, iteration 2) was a zero-API stress test. On the union panel (362 de-duplicated episodes, 54 concepts), ΔAUC over M2 was $+0.001$ $[-0.012, 0.012]$. A shuffled- R placebo’s 95th percentile (0.130) exceeded the original $+0.103$, so the iteration-1 lead could not be certified as above chance on 80 episodes. All G-variant O1 gains were label-coverage artefacts (G : $+0.072 \rightarrow +0.002$).

3.5 Dataset 2: When research concepts were officially recognised

Dataset 2 (`gen_art_dataset_2`, iteration 2) built an external-recognition lookup table (outcome O_5) for all 65,026 OpenAlex legacy concepts. Sources: MeSH 2026 (20,872 concepts), English Wikipedia creation dates (6,540 exact; 50,459 year-usable), Wikidata P571/P575 (1,425 concepts), ACM CCS, MSC, PACS/PhySH taxonomy events, and curated lists (Nature Methods MoTY, Science BOTY, Physics World BOTY, MIT TR10, Gartner Hype Cycle, Clarivate/CAS Research Fronts). O_5 was later found to be unrelated to publication outcomes (pooled ρ with $O_{2r} = 0.014$ $[-0.045, 0.073]$; 67% of concepts are recognised at or before t_0 ; only 42% of positives mark a genuinely new concept).

3.6 Research 1: How our results compare with related papers

Research 1 (`gen_art_research_1`, iteration 2) produced a positioning study. Key comparisons: Guevara 2016 field-entry AUC 0.896/0.715/0.682 (individuals/organisations/countries); Weng et al. 2013 community diversity predicts virality; Cheng et al. 2023 ideational consistency predicts volume. The Applied Network Science “Networks for everyday life” collection could not be fully accessed.

3.7 Dead ends from iteration 2

- Gateway centrality for field retention: **disconfirmed** ($\Delta\text{AUC} -0.00001$).
- Gateway weighting for field entry: adds nothing beyond plain retaining relatedness.
- Two-class trajectory typology: HMM-vs-DTW ARI = 0.094.
- Ordering (gateway-first): **mixed**—negative lead-lag, pre-trend, reverse path significant on DEV.
- O1 gains of G variants: label-coverage artefacts.
- O_5 external recognition: unrelated to publication outcomes.

3.8 Review and hypothesis update

The reviewer scored iteration 2 at 3 (blocking). Main issue: stated conclusions contradicted artifacts (H1 “disconfirmed by all criteria” but the within-field LPM passed; ordering listed as confirmed despite negative lead-lag evidence). The hypothesis moved to **deepen** the retained-frontier lead ($d_0 = +0.28$, one frame). Evidence state: *lead*. Confidence: *decreased*.

4 Iteration 3: The retained frontier and the indicator screen

4.1 Strategy and rationale

Iteration 3 had three goals: (1) a decisive test of the retained-frontier claim against the field-standard RCA-thresholded density; (2) the RQ1 held-out indicator screen (53 indicators, 6 families); (3) record repair. Five artifacts were commissioned; four completed. The RQ2 trajectory artifact (`gen_art_experiment_9`) failed (output-format validation failed after 5 retries; `method.py` never ran).

4.2 Experiment 7: Do concepts spread from fields that keep them?

Experiment 7 (`gen_art_experiment_7`, iteration 3) ran the decisive retained-frontier test on an independent frame (Experiment 5 minus every Experiment 6 concept: 11,841 concepts; DEV 4,486; held-out 3,162 concepts with 6,978 entry events in 6,076 informative strata). The spec was hash-frozen on DEV and scored once on held-out.

Table 5: Retained-frontier conditional-logit results (Experiment 7, iteration 3). Source: `step2_heldout.json`.

Body			d_0	95% CI	LR (R3 vs R2)	n_{concepts}
Held-out	pooled	(PHYS+LIFEENV+SOC+MATHDEC)	+0.322	[0.291, 0.355]	325.8	3,162
Physical Sciences			+0.148	[0.074, 0.219]	—	656
Life & Environment			+0.401	[0.347, 0.458]	—	1,071
Social Sciences			+0.297	[0.245, 0.345]	—	1,274
Math & Decision (underpowered)			+0.065	[−0.110, 0.234]	—	161
Cohort 2010–14			+0.321	[0.292, 0.347]	—	—
DL pooled (4 groups)			+0.243	[0.118, 0.368]	—	—

d_0 survives the $\text{RCA} > 1$ density and volume-weighted density ladders (S_{strict} : 0.304 [0.268, 0.336]). Specificity tests reject their nulls after Holm correction (retained-label permutation $p = 0.001$, rewired backbone $p = 0.004$, node-label $p = 0.003$).

Dose–response by retention age. DEV: +0.056 (2 yr), +0.103 (3 yr), +0.251 (≥ 4 yr)—monotone. Held-out: +0.098 / +0.075 / +0.304—*not* monotone (`monotone_nondecreasing` = false, Spearman = 0.5).

Frozen verdict: FRONTIER = PARTIAL. The pre-declared volume-matched contrast (retained vs entered-not-retained in the same current $\times$ cumulative volume cell) is null: -0.028 [−0.105, 0.046] held-out (−0.008 DEV). Under Hidalgo min-conditional-probability proximity, $d_0 = -0.021$ ($p = 0.012$): the effect is backbone-specific.

4.3 Experiment 8: Which early network signals travel across fields

Experiment 8 (`gen_art_experiment_8`, iteration 3) ran the RQ1 held-out deliverable: 53 indicators in 6 families over $t_0..t_0 + 2$ against the B5 baseline, on the 12,499-concept Experiment 5 frame. Top 10 per outcome frozen on DEV, scored once on held-out with DL pooling and Holm correction.

Seven of the ten are confirmed, spanning two families: retained frontier (FR: M0.density_end, D_vol_end, CONTACT_REACH, RETENTION_RATIO) and co-occurrence ego-network topology

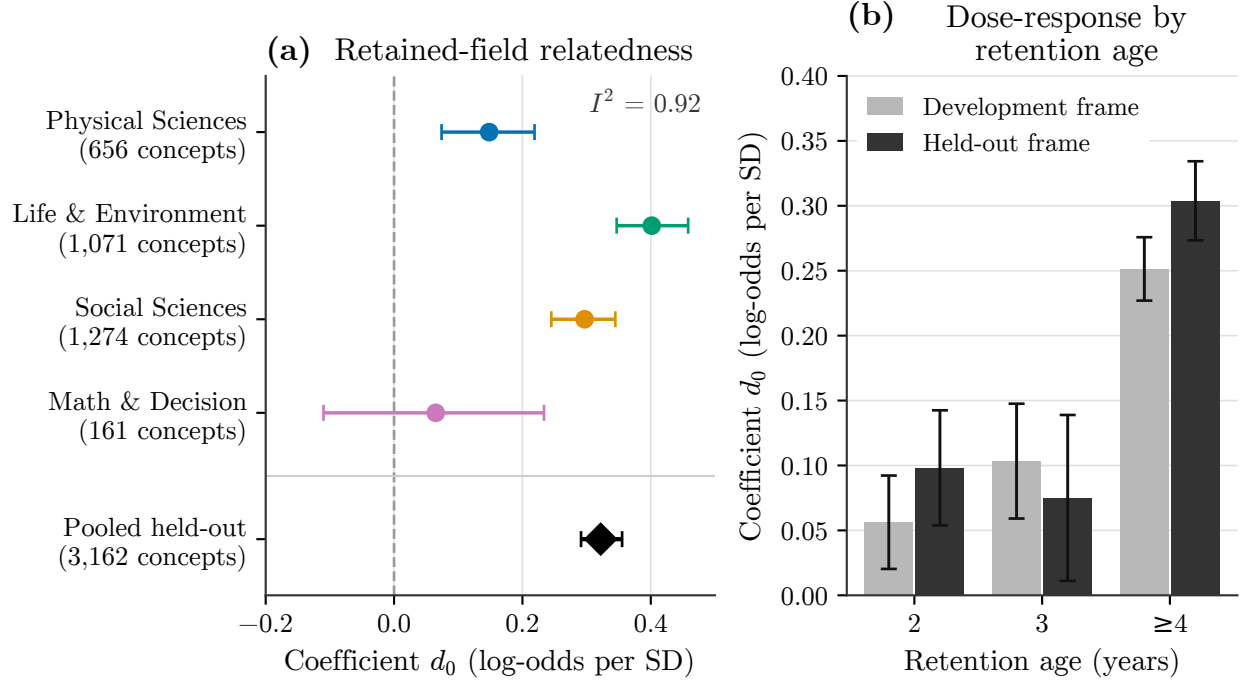


Figure 3: Retained-field relatedness predicts the next field a concept enters. (a) Conditional-logit coefficient d_0 on the held-out groups of the independent frame. Coloured circles are per-group estimates with 95% CIs. The black diamond is the pooled held-out estimate, $d_0 = 0.322$ [0.291, 0.355]. Three groups are clearly positive; the Math & Decision interval crosses zero ($I^2 = 0.92$). (b) d_0 split by retention age, for the development frame (light grey) and the held-out frame (dark grey). In both frames the signal is largest for fields that have retained the concept for ≥ 4 years.

Table 6: Held-out indicator screen: the 10 indicators frozen on DEV for rarefied breadth (O_{2r} , $m = 50$), tested on held-out groups. DerSimonian–Laird pooled estimates with 95% CI. Source: `heldout.summary.json` (Experiment 8, iteration 3).

Indicator	Family	Pooled PSP	95% CI	I^2	Sign	Confirmed
M0_density_end	FR	+0.375	[+0.279, +0.462]	0.74	6/6	Yes
D_vol_end	FR	+0.307	[+0.256, +0.356]	0.10	6/6	Yes
CONTACT_REACH	FR	+0.211	[+0.161, +0.261]	0.00	6/6	Yes
n_{comm} (W3)	A	+0.167	[+0.063, +0.267]	0.78	6/6	Yes
NOV _{res}	A	+0.151	[+0.044, +0.255]	0.75	6/6	Yes
RETENTION_RATIO	FR	−0.114	[−0.160, −0.067]	0.00	6/6	Yes
ego_density (W3)	A	−0.102	[−0.151, −0.053]	0.00	6/6	Yes
Rao–Stirling	A	−0.072	[−0.153, +0.010]	0.44	5/6	No
G_{btw}	G	+0.056	[−0.006, +0.118]	0.33	6/6	No
log offhome vol.	F	−0.089	[−0.171, −0.007]	0.63	5/6	No

(A: n_{comm} , NOV_{res} , ego_density). M0_density_end and D_vol_end use cumulative field history up to $t_0 + 2$, so part of their signal is a pre-onset footprint.

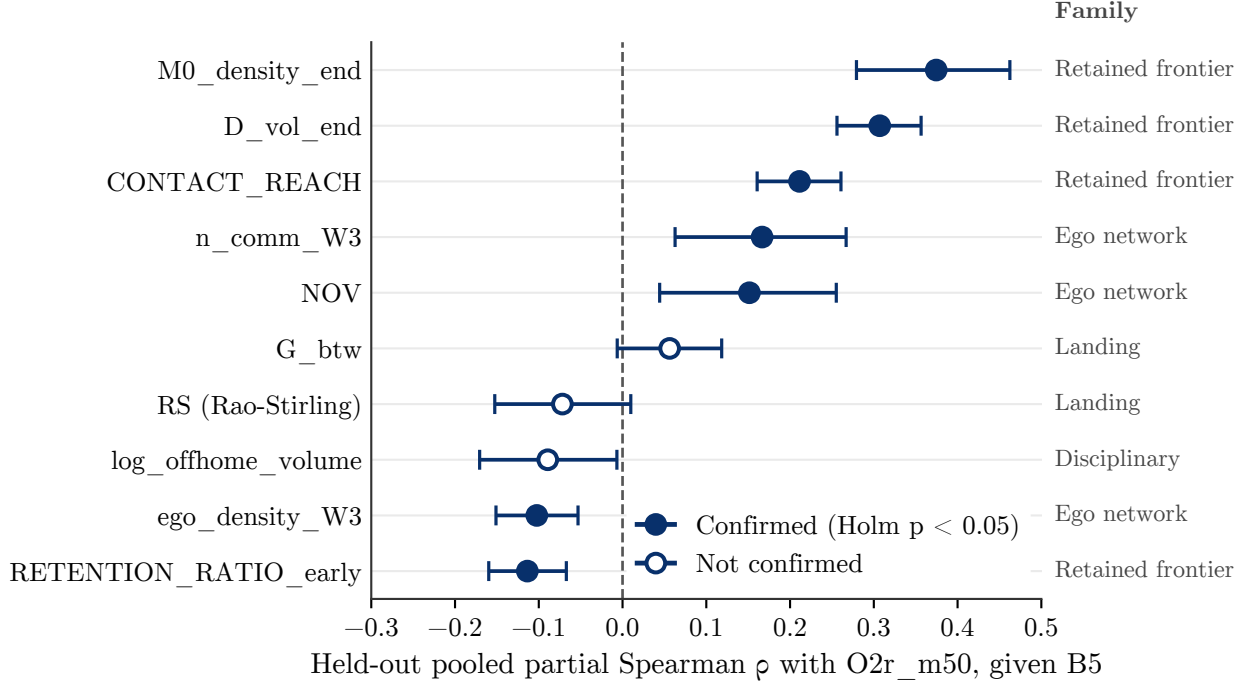


Figure 4: Held-out screen of the 10 indicators frozen on DEV for rarefied cross-field breadth (O2r_m50). Each row shows the DerSimonian–Laird pooled partial Spearman ρ with O2r_m50 given the B5 baseline. Horizontal lines are 95% CIs. Filled circles mark confirmed indicators (Holm $p < 0.05$); open circles mark unconfirmed indicators. Seven of the ten are confirmed: five positive and two negative.

An ElasticNet combining all indicators adds Spearman $+0.059$ [$+0.046$, $+0.073$] over B5 on held-out groups; an Explainable Boosting Machine adds $+0.052$ [$+0.037$, $+0.067$].

Other outcomes. O_1 (sustained uptake): only $n_{\text{authors_early}}$ confirmed ($+0.161$). O_4 (citation growth): REL_home -0.114 , author_growth $+0.065$; EBM Spearman 0.188 vs B5 0.015 . O_3 (transience): $n_{\text{authors_early}}$ is the only confirmed indicator ($+0.089$ [0.031 , 0.148]); L1-logit gains $+0.093$ AUC over a B5 at chance (0.506). $O_5/O_{5,\text{WW}}$: no indicator or model beats B5 + onset year.

4.4 Evaluation 2: Auditing the record before the paper

Evaluation 2 (`gen_art_evaluation_2`, iteration 3) audited 246 claims: 224 MATCH, 15 MISLABELLED, 6 MISMATCH. It produced correction blocks and record tables (portability, partial associations, H1 criteria, ordering). The cross-frame agreement showed onset exact agreement 0.976 , home kappa 0.99 , O2r $\rho = 0.998$, but retention kappa only 0.28 (0.98 with the matched R_{abs2} rule).

4.5 Research 2: Prior art and venue check

Research 2 (`gen_art_research_2`, iteration 3) checked the novelty of the retained-frontier claim: PARTIALLY ANTICIPATED. No prior study combined a persistence-filtered relatedness *predictor*

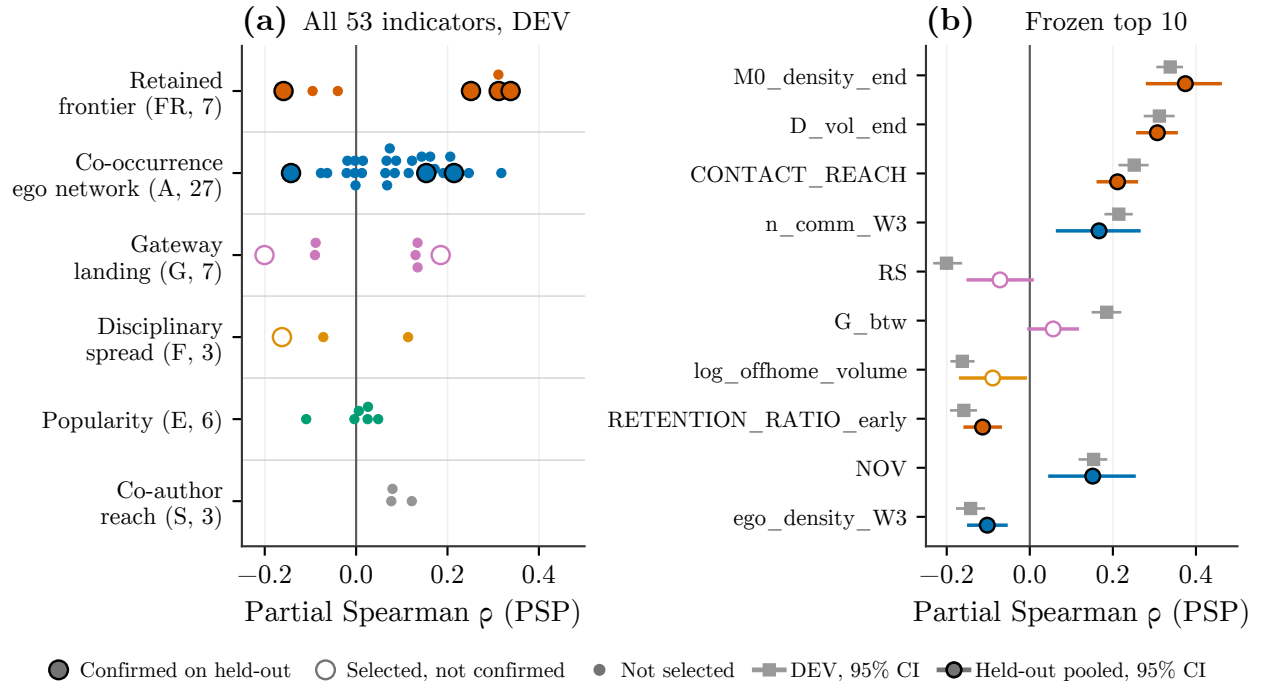


Figure 5: Complete indicator screen for rarefied cross-field breadth (O2r). (a) Partial Spearman ρ of all 53 screened indicators on DEV, colour-coded by family. Large filled circles are the 7 confirmed; large open circles are the 3 selected but not confirmed; small dots are the 43 not selected. (b) The frozen top 10 in DEV rank order. Grey squares are DEV estimates; circles are held-out DL-pooled estimates.

for entry; Pinheiro et al. 2022, Albora et al. 2023 and Bahar et al. 2014 use persistence only as an outcome filter. Research 2 also noted a missing rival: $D_{\text{rca_persist_k}}$, which was later found to differ from Experiment 7’s $D_{\text{rca_pers}}$ (max $\rho = 0.877$).

4.6 Failed artifact: Experiment 9

`gen_art_experiment_9` (“How new concepts spread: paths and reasons”) was commissioned for RQ2 trajectories, decomposition and case studies on the 12,499-concept panel. It failed: output-format validation failed after 5 retries; `method.py` never ran.

4.7 Dead ends from iteration 3

- Gateway centrality for field retention: confirmed as dead on the full panel.
- Rao–Stirling, G_{btw} , log offhome volume: not confirmed on held-out.
- O_5 external recognition: no indicator beats B5.
- Experiment 9 (RQ2 trajectories on large panel): failed, not run.
- Candidate S (co-author reach): tested in Experiment 8 (S_{comp} , $S_{\text{comp_n}}$, $S_{\text{isolated_share}}$); none confirmed.

4.8 Review and hypothesis update

The reviewer scored iteration 3 at 3 (blocking). Main issues: O_4 citation-growth results mislabelled as transience; preregistered predictions misstated; Evaluation 2 corrections not applied. The hypothesis moved to **deepen** the Experiment 8 lead (openness vs consolidation). Evidence state: *lead*. Confidence: *decreased*.

5 Iteration 4: Fresh-cohort confirmation and RQ2 trajectories

5.1 Strategy and rationale

Iteration 4 attacked the Experiment 8 openness lead on four sides: (1) replication on a never-screened 2015–2017 onset cohort; (2) confound controls (concept type, pre-onset footprint, mechanical coupling via a home-only build); (3) RQ2 decomposition and trajectories on the full 12,499-concept panel; (4) record repair and boundary analysis. Five artifacts were commissioned; four completed plus one incomplete.

5.2 Experiment 10: Fresh-cohort confirmation of OPEN

Experiment 10 (`gen_art_experiment_10`, iteration 4) tested OPEN on a fresh 2015–2017 onset cohort (1,443 concepts after LLM precision gate; 634 with O_{2r} $m = 50$; 573 with $\text{OPEN}_{\text{home}}$) that no earlier screen had touched. The spec was hash-sealed before the unseal.

Frozen verdict: confirmed but marginal. $\text{OPEN}_{\text{home}}$ PSP = +0.091 [+0.013, +0.171] at R2 and +0.080 [+0.001, +0.162] at R3. CIs include zero at R4/R5. DL pool over groups is +0.083 [−0.007, +0.173]. Holm $p = 0.048$. $\text{OPEN}_{\text{home}}$ adds no practical prediction (+0.002 [−0.003, +0.008] over B5). Mechanical coupling is large: OPEN_{all} minus $\text{OPEN}_{\text{home}}$ is +0.093 [+0.016, +0.169]. Pre-seal power was only 0.16.

Table 7: OPEN index control ladder on the 2015–2017 confirmatory cohort. PSP = partial Spearman with rarefied breadth (O_{2r} , $m = 50$), 95% concept-bootstrap CI. Source: `cohort_result.json` (Experiment 10, iteration 4).

Build	R0	R1	R2	R3	R4	R5	n
OPEN _{home}	+.12	+.10	+.09	+.08	+.07	+.06	573
OPEN _{all}	+.21	+.18	+.17	+.17	+.15	+.14	630
OPEN _{size}	+.18	+.15	+.15	+.14	+.12	+.11	591

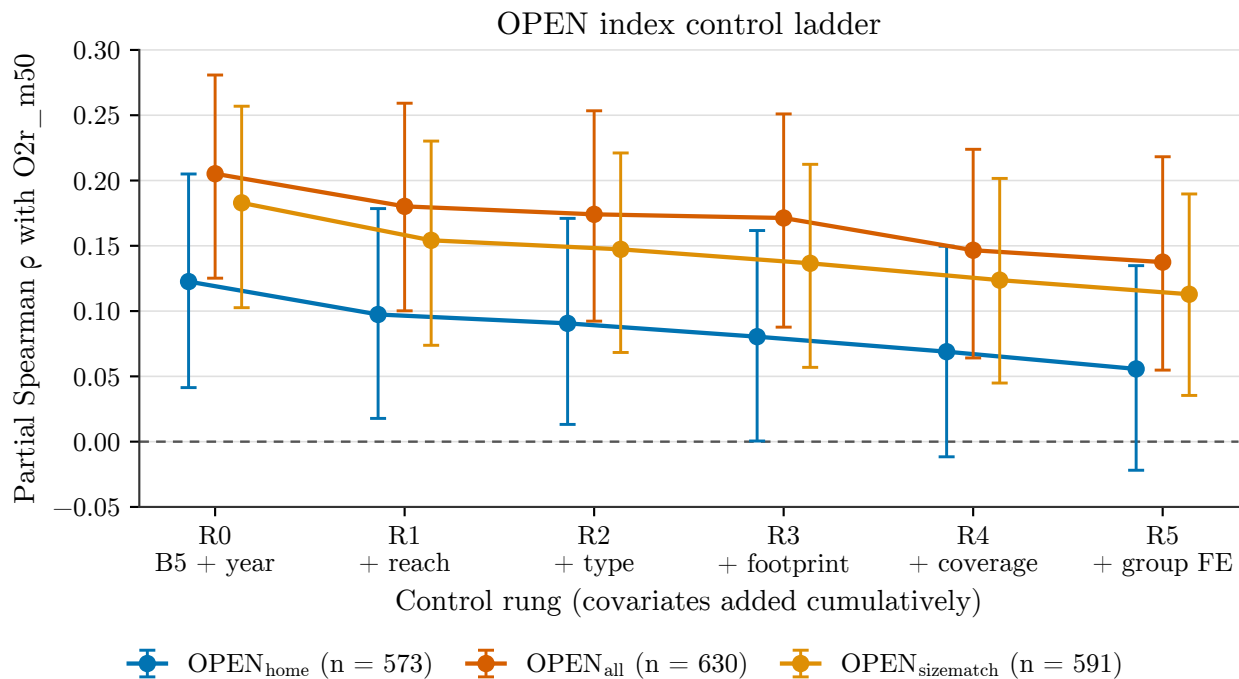


Figure 6: Control ladder for the OPEN index on the fresh 2015–2017 onset cohort. Circles are point estimates; error bars are 95% CIs ($B = 2,000$). OPEN_{all} and OPEN_{size} stay above zero at every rung, but OPEN_{all} is partly coupled mechanically. The home-only build is marginal: its CI touches zero at R3 and includes zero at R4 and R5.

5.3 Experiment 11: Within-concept closure test (incomplete)

Experiment 11 (`gen_art_experiment_11`, iteration 4) sealed a within-concept pre-registration and ran DEV body models on 35,328 concept-years from 4,661 concepts. Both preregistered effects are null on DEV: density PPML $b = -0.070$ $[-0.180, 0.040]$; $\text{OPEN}_{\text{home}} b = +0.015$ $[-0.038, 0.069]$. DL density: -0.075 $[-0.210, 0.061]$. By the frozen rule: **NOT SUPPORTED**. The Sun–Abraham event study was interrupted (no worker result). Held-out and cohort were not run in this iteration.

5.4 Experiment 12: Breadth decomposition and trajectories

Experiment 12 (`gen_art_experiment_12`, iteration 4) ran the RQ2 analysis on all 12,499 concepts.

Decomposition. The log-additive decomposition of the top-vs-bottom $O_{2r,\text{resid}}$ tercile gap in retained off-home breadth:

Table 8: Breadth decomposition ($\log B_n = \log E_2 + \log M + \log \rho$): share of the tercile gap attributable to each component. Source: `decomposition_dev.json`, variant `i_pooled` (Experiment 12, iteration 4). $n = 3,188$ DEV concepts.

Component	Share	95% CI
Exploration ($s_{E2} + s_M$)	0.732	[0.703, 0.764]
Early contact diversity (s_{E2})	0.779	[0.738, 0.818]
Frontier advance (s_M)	-0.046	$[-0.074, -0.017]$
Retention (s_ρ)	0.268	[0.236, 0.297]
Difference (explore – retain)	0.464	[0.407, 0.528]

The preregistered PR1 variant (iv, Medicine excluded) gives DEV 0.633 [0.537, 0.727], held-out pooled 0.492 [0.403, 0.575], cohort 0.445 [0.358, 0.527], DL 0.504 [0.329, 0.679], $I^2 = 0.76$. Broad concepts start with wider contact; the frontier advance component is near zero or negative.

Trajectory typology. No discrete typology passes the naming rule: DTW $k = 4$ vs HMM $S = 5$ ARI = 0.222; Hennig Jaccard 0.69–0.82; held-out re-cluster ARI 0.44/0.38. Verdict: **CONTINUUM**. PC1 (38.8%) is the breadth-of-spread axis, PC2 (10.7%) is the keep-vs-lose axis. OPEN tracks PC1, not PC2.

Sequence test. No ordering signal beyond the mechanical lag (excess ≤ 1.7 percentage points, sign flips by body). Intersection-born concepts take off off-home later (HR ~ 0.47 [0.42, 0.54]).

5.5 Evaluation 3: Record fixes and openness robustness

Evaluation 3 (`gen_art_evaluation_3`, iteration 4) produced corrections and boundary results.

Pre-onset footprint. About half of the M0_density_end and D_vol_end signal is pre-onset: M0_density_end drops from +0.374 to +0.187 post-onset (attenuation 0.50 [0.38, 0.60]); D_vol_end from +0.317 to +0.176 (0.45). Post-onset D_vol is nearly rank-identical to B5 reach ($\rho = 0.97$ –1.00).

Specification curve. 1,920 specifications (120 composites $\times$ 4 outcomes $\times$ 4 control sets): 99.7% of pooled CIs above zero; all estimates positive; median 0.152; Freedman–Lane $p = 0.005$. This is exploratory (all-papers build, previously unsealed groups).

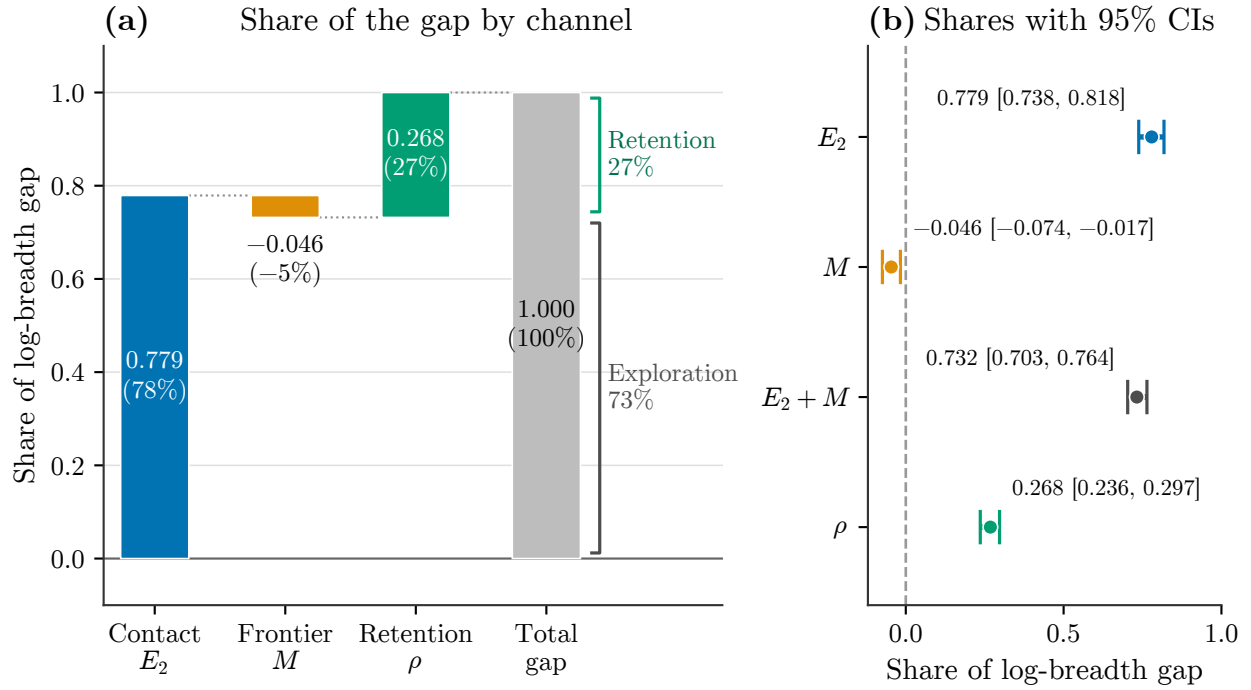


Figure 7: Log-additive decomposition of the gap in retained off-home breadth between the top and bottom terciles of size-adjusted breadth. DEV concepts, $n = 3,188$. (a) Waterfall: early contact diversity E_2 (blue) contributes 0.779 (78%), frontier advance M (amber) is slightly negative, and retention ρ (green) contributes 0.268 (27%). The bracket groups $E_2 + M$ as exploration (73%) and ρ as retention (27%).

Claims ledger. Version 3: 1,290 rows, 0 MISMATCH, 0 NOT_FOUND. An independent verifier reproduced every row.

5.6 Research 3: Novelty check

Research 3 (`gen_art_research_3`, iteration 4) checked prior art. Openness $\rightarrow$ breadth (C1): **partially anticipated**—Maillart 2026, Wang 2017, Weng 2013, Ugander 2012 show the same direction for different units. Consolidation $\rightarrow$ less breadth (C2): **partially anticipated** in mechanism, **contradicted** on other outcomes (Cheng et al. 2023: consistency $\rightarrow$ volume; Centola 2010 and Romero 2011: clustering helps adoption). Low retention ratio (C3): **new**. Within-concept closure (C4): **new as a lead-lag test**, though Experiment 11 was null on DEV.

5.7 Dead ends from iteration 4

- Experiment 11 within-concept closure: NOT SUPPORTED on DEV.
- Trajectory typology: CONTINUUM (no naming-rule pass).
- PR2 (localised keep more early): REVERSED on DEV (-0.110) and cohort (-0.058).
- RETENTION_RATIO_early: does not survive concept-type controls on the cohort (R2 -0.043 [$-0.116, 0.031$]).
- Sequence ordering: no signal beyond mechanical lag.

5.8 Review and hypothesis update

The reviewer scored iteration 4 at 2 (blocking). Key issues: fabricated case-study rows in an early draft; Experiment 11 absent; OPEN conclusions contradicted Experiment 10’s own reading; several corrections still unapplied. The hypothesis moved to **deepen** the lead: confirm the decoupled churn/novelty signal on a vocabulary-free second population. Evidence state: *lead*. Confidence: *decreased*.

6 Iteration 5: Vocabulary-free confirmation and confound analysis

6.1 Strategy and rationale

The final iteration had five goals: (1) a sealed confirmation on vocabulary-free “Frame N” concepts (brand-new phrases absent from all legacy vocabularies); (2) the Cheng consistency–breadth reversal test; (3) a mechanism analysis (which partners carry the signal) and completion of the Experiment 11 closure test; (4) a confound analysis of whether churn is real or thin-sample noise; (5) evidence synthesis and record repair.

6.2 Experiment 13: Does the churn signal hold for brand-new phrases?

Experiment 13 (`gen_art_experiment_13`, iteration 5) mined 636 vocabulary-free phrase-born concepts from random title samples (2003–2015 onsets), excluding all legacy labels and previously scored concepts. These Frame N newborns have 14% lower breadth and 89% higher transience than legacy concepts, confirming that curated vocabularies are survivor-selected.

Frozen verdict: PARTIAL. $\text{OPEN}_{\text{home}}$ PSP = $+0.117$ [$+0.020$, $+0.218$] at R3 on O_{2r} ($m = 30$); $+0.086$ [-0.009 , $+0.190$] at R5 (CI includes 0). On O_{2r} ($m = 50$): $+0.161$ and $+0.122$, both CIs above 0. $\text{NOVCHURN}_{\text{home}}$ at R3: $+0.108$ [$+0.007$, $+0.211$]. 3 of 4 estimable groups positive (SOC -0.025); DL $+0.112$ [-0.015 , $+0.239$]; Holm $p = 0.052$. Pre-unseal power was 0.47.

NOV_{res} carries the signal ($+0.208$ [$+0.113$, $+0.303$]); edge persistence is null (-0.013). Coupling (ALL-HOME = $+0.056$) is not significant. The Cheng reversal is NOT confirmed on Frame N (-0.064 [-0.159 , 0.031]). No forecasting gain over B5 (Spearman 0.80).

Pooling Frame N with the Experiment 10 legacy cohort (exploratory, different rungs): $+0.096$ [$+0.034$, $+0.158$] at R3.

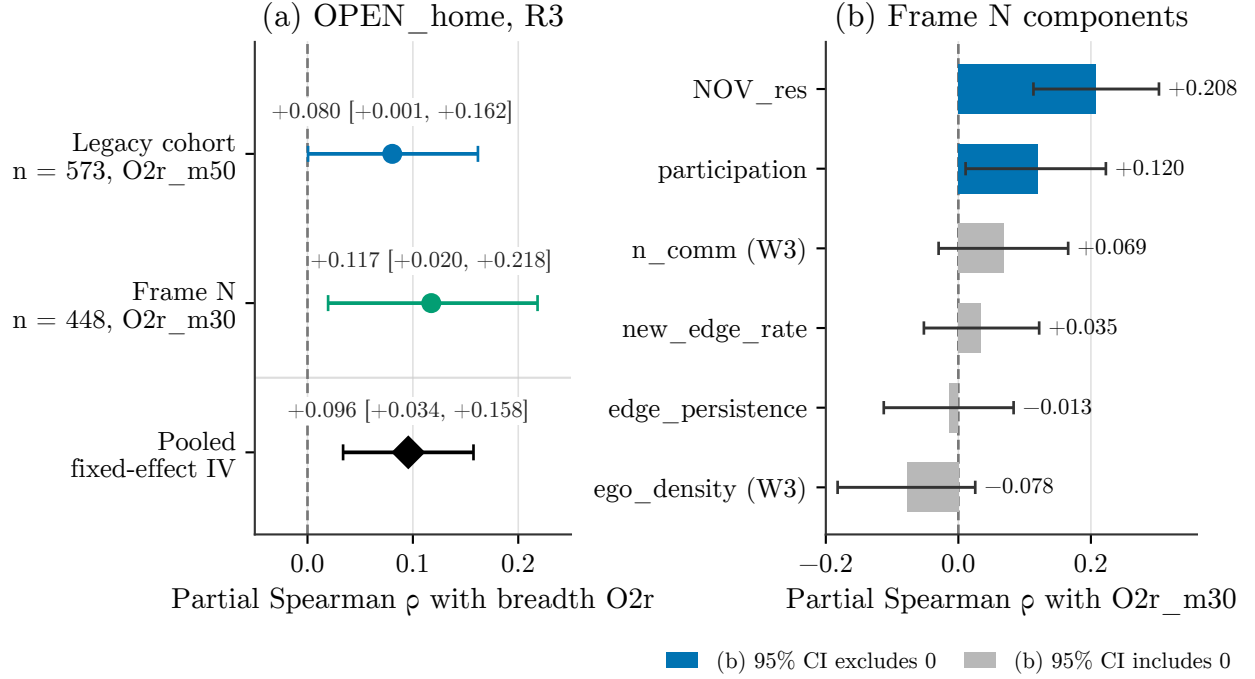


Figure 8: Confirmation of the OPEN signal on vocabulary-free Frame N concepts. (a) Forest plot of $\text{OPEN}_{\text{home}}$ PSP with rarefied breadth at R3: the 2015–17 legacy cohort ($n = 573$, $+0.080$), Frame N ($n = 448$, $+0.117$), and their exploratory pool ($+0.096$ [$+0.034$, $+0.158$]). (b) PSP for the six $\text{OPEN}_{\text{home}}$ components on Frame N. NOV_{res} carries the signal ($+0.208$); edge persistence is null (-0.013).

6.3 Experiment 14: Cheng’s consistency—a size effect, not a reach predictor

Experiment 14 (`gen_art_experiment_14`, iteration 5) rebuilt Cheng et al.’s (2023) ideational consistency for 12,311 concepts (105,839 concept-years).

Adding current volume $\log V(t)$ removes about 98% of Cheng’s effect. As an early trait, consistency predicts *narrower* cross-field reach (-0.069). The reversal holds in all five field groups ($I^2 = 0$), replicates on the 2015–17 cohort (-0.111), but is NOT confirmed on Frame N. Within concepts, a more consistent year is followed by slightly more entries ($+0.025$)—the reach penalty is a between-concept trait. Consistency has Spearman $+0.77$ with edge persistence: it is essentially weighted persistence.

Table 9: Cheng’s ideational consistency: volume vs breadth (Experiment 14, iteration 5). Source: `cheng_verdict.json`.

Test	Estimate	CI / detail
<i>Test A: Next-year volume (Cheng’s DV)</i>		
NB twin of Cheng	$b = 0.428$ (+53.5%/SD)	—
PPML	+83.1%	[+71%, +97%]
PPML + $\log V(t)$	+1.3%	[+0.5%, +2.1%]
A2/A1 ratio	0.021	[0.009, 0.035]
<i>Test B: Early trait PSP with O_{2r} ($m = 50$) given B5</i>		
Pooled (5 groups)	−0.069	[−0.093, −0.047]
DL (5 groups)	−0.079	$I^2 = 0$, 5/5 negative
2015–17 cohort replication	−0.111	[−0.197, −0.030]
<i>Test C: Within-concept</i>		
Consistent yr $\rightarrow$ entries	$b = +0.025$	[0.004, 0.048]

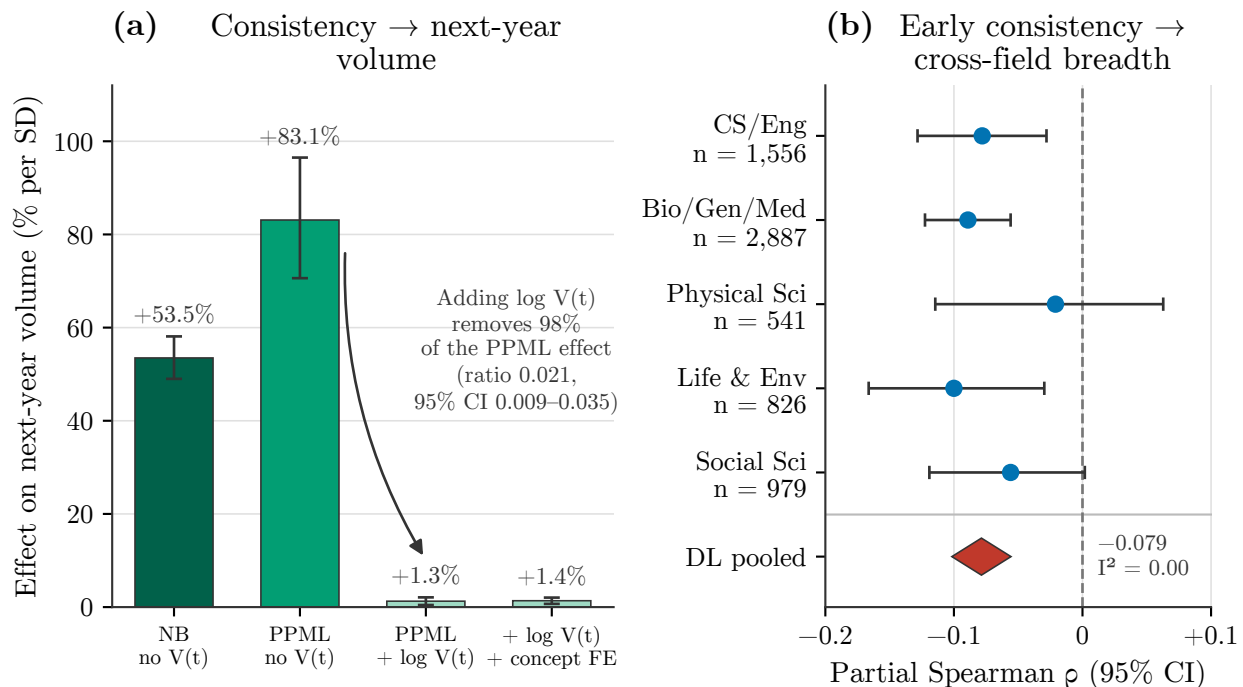


Figure 9: The consistency–breadth reversal. (a) Effect of ideational consistency on next-year volume. Adding $\log V(t)$ removes about 98% of the effect (+53.5% $\rightarrow$ +1.3%). (b) Partial Spearman between early consistency and rarefied breadth given B5. All five field groups are negative; DL pooled −0.079 ($I^2 = 0$).

6.4 Experiment 15: Why churning concepts spread; Experiment 11 completion

Experiment 15 (`gen_art_experiment_15`, iteration 5) had three parts.

Part A: Partner decomposition. $\text{NOVCHURN}_{\text{home}}$ replicates: pooled $+0.118$, held-out DL $+0.097$ ($I^2 = 0$), cohort $+0.171/+0.144$ at R0/R3. The signal comes from new-community partners ($C_2 = +0.102$, Holm $p = 0.0025$) that arrive through mixed-field papers ($C_4 = +0.103$). Domain partners carry the signal (Shapley $\phi = +0.132$ on the 2015–17 cohort), not methodological partners ($+0.012$). Bridging papers (5% of early home papers) halve NOVCHURN ’s PSP from 0.118 to 0.056.

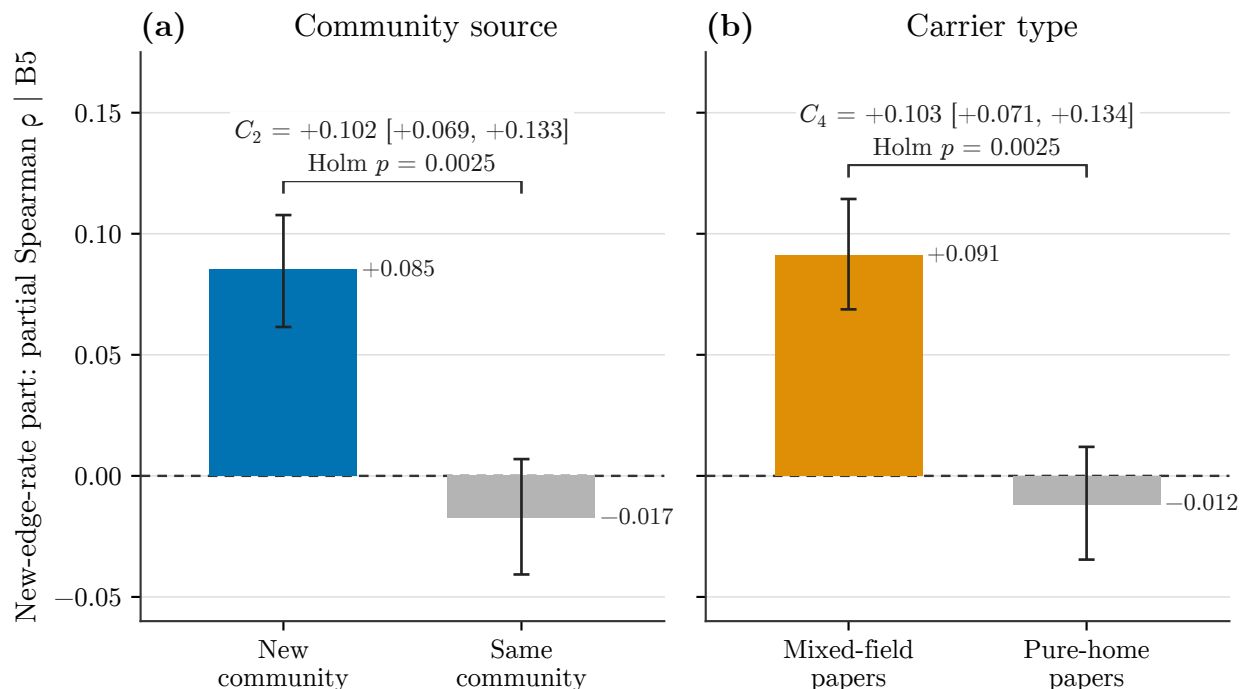


Figure 10: Which co-occurrence partners carry the openness–breadth signal? (a) Partners from new Leiden communities (blue, $+0.085$) carry the signal; same-community partners (grey, -0.017) do not ($C_2 = +0.102$, Holm $p = 0.0025$). (b) Partners arriving through mixed-field papers (orange, $+0.091$) carry it; pure-home papers (grey, -0.012) do not ($C_4 = +0.103$, Holm $p = 0.0025$).

Part B: Trait stability. Yearly $\text{OPEN}_{\text{home}}$ ICC is 0.34–0.39 (NOVCHURN 0.26–0.29). Window retest is 0.51–0.57; disattenuated retest 0.86–0.91. Openness is a fair between-concept trait measured through a noisy yearly window.

Part C: Experiment 11 completion. The sealed Experiment 11 closure test was completed from cached data. DEV verdict unchanged: NOT SUPPORTED. On `OLD_HELDOUT`, $\text{OPEN}_{\text{home}}$ is -0.079 [$-0.146, -0.013$]*—the opposite of the predicted sign*. COHORT both null. Sun–Abraham event study (DEV, never-treated): lag 0.2 = -0.018 [$-0.042, +0.004$], pre-trend $p = 0.52$, event-date placebo $p = 0.19$; held-out and cohort null. H-S1 (post-prominence half-life) holds on DEV ($+0.113$), COHORT ($+0.105$) and pooled ($+0.076$ [$+0.024, +0.126$]) but not `OLD_HELDOUT` ($+0.001$). H-P1 as preregistered fails: the community half is $+0.216$ [$+0.081, +0.351$], the method half -0.055 .

6.5 Evaluation 4: Record repair and openness evidence pool

Evaluation 4 (`gen_art_evaluation_4`, iteration 5) cleared all 10 MUST-FIX items and produced the corrected report.

Evidence synthesis (descriptive). $\text{OPEN}_{\text{home}}$ over 6 non-selection bodies (4 held-out groups + 2010–14 cohort + 2015–17 cohort): DL +0.069 [+0.038, +0.100], HKSJ [+0.042, +0.096], $I^2 = 0$, 6/6 positive. DEV selection body: +0.109, shrinkage ratio 1.58. $\text{NOVCHURN}_{\text{home}}$ ($k = 5$): +0.105 [+0.069, +0.140]. Shuffled-feature placebo pool: +0.021 [−0.010, +0.051] (includes 0).

Claims ledger v4. 1,769 rows, 0 MISMATCH, 0 NOT_FOUND, 0 orphans. All values present in their target sections.

6.6 Experiment 16: Is neighbourhood churn real or thin-sample noise?

Experiment 16 (`gen_art_experiment_16`, iteration 5) ran confound checks on 13,444 selection concepts.

Table 10: Confound-corrected NOVCHURN variants (Experiment 16, iteration 5). Pooled PSP with O_{2r} ($m = 50$) given B5. Source: `clean_vs_raw_psp.json`.

Variant	Pooled PSP	95% CI	Retention ratio
$\text{NOVCHURN}_{\text{raw}}$	+0.116	[+0.09, +0.14]	1.00
V1: fixed- n rarefaction ($n = 10$)	+0.078	—	0.68
V2: within-concept permutation excess	+0.008	[−0.02, +0.03]	0.065
V3: configuration-null z -score	+0.106	—	1.00
Chao-2005 Jaccard	—	—	0.91
$\text{OPEN}_{\text{home, clean}}$ (cfg z -scores)	+0.115	—	1.25
$\text{OPEN}_{\text{home}}$ (raw)	+0.092	—	1.00

Key finding. The within-concept year-label permutation excess is indistinguishable from zero (+0.008), cutting NOVCHURN from +0.116 to near zero. Raw persistence is 66% explained by its own V2 null mean (thin-sample share), and its correlation with $\log n$ is +0.72. What predicts breadth is a static property—the non-redundancy of the concept’s home-topic partner mix—not year-to-year turnover. However, V2 excess variants have split-half SB ~ 0.01 – 0.05 and the planted-churn check (PC2) fails, so V2 cannot adjudicate temporal churn at ~ 10 papers/year.

Split-half reliability of $\text{NOVCHURN}_{\text{raw}}$ is 0.48, rising to 0.67 at ≥ 100 home papers. Disattenuated pooled $\text{NOVCHURN}_{\text{raw}}$ PSP: 0.178 [0.141, 0.214] (approximate; CI not independently re-derived).

6.7 Dead ends from iteration 5

- Frame N Cheng reversal: not confirmed (−0.064, CI includes 0).
- Frame N $\text{OPEN}_{\text{home}}$ at R5: CI includes 0 (+0.086 [−0.009, +0.190]).
- Frame N Holm $p = 0.052$: does not survive strict correction.

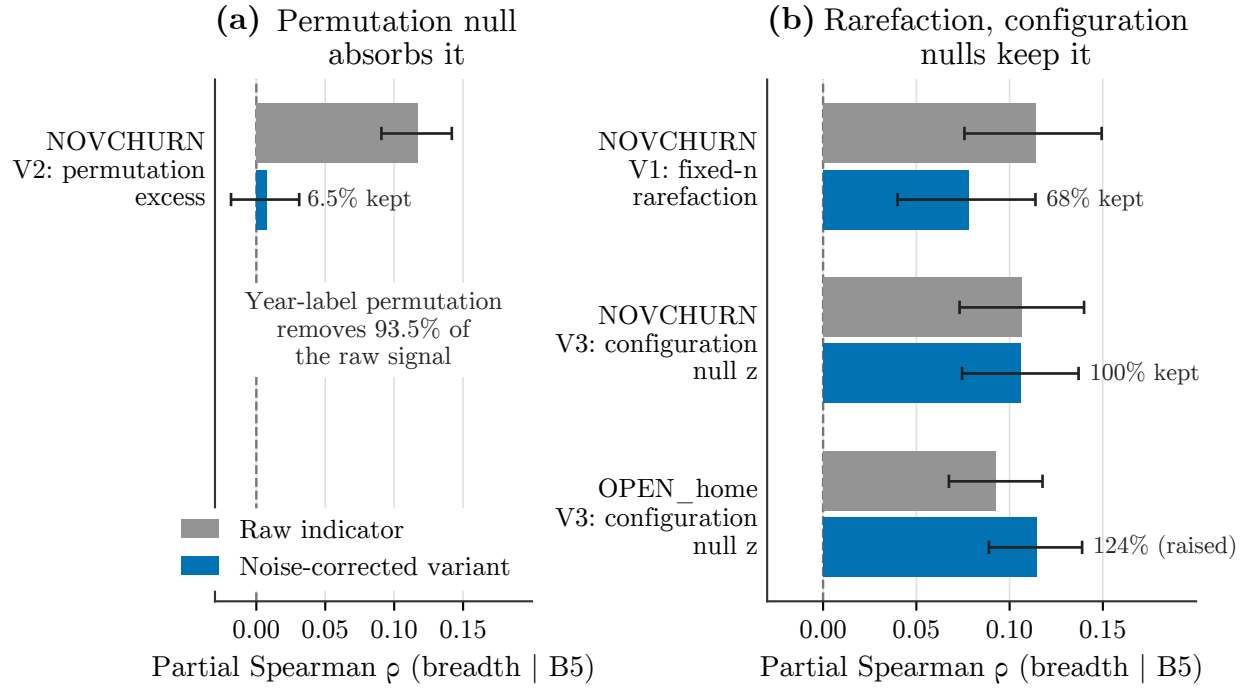


Figure 11: The churn/openness signal behaves like topical non-redundancy, not year-to-year partner turnover. (a) The within-concept year-label permutation excess (V2) cuts NOVCHURN from +0.117 to +0.008, keeping 6.5%. (b) Fixed- n rarefaction keeps 68%; the configuration-null z -score keeps 100% and raises OPEN_{home} to 124%.

- Experiment 11 closure: NOT SUPPORTED; opposite sign on OLD_HELDOUT.
- V2 excess churn: cannot adjudicate temporal turnover (too unreliable).
- Trait prediction P-B1: fails (yearly $\text{OPEN}_{\text{home}}$ ICC = 0.34–0.39).
- METHOD excess of partner decomposition: DEV-only, not replicated on cohort.
- No forecasting gain from early network structure on any body.

6.8 Review and hypothesis update

The final reviewer scored iteration 5 at 2 (blocking). The submitted document was the paper manuscript rather than the research notebook; the reviewer required the chronological record. Evidence state: *lead*. The headline measurement from Evaluation 4 is the descriptive pool of $\text{OPEN}_{\text{home}}$ over 6 non-selection bodies: PSP +0.069 [+0.038, +0.100], $I^2 = 0$, 6/6 positive.

7 What the run learned and what is still open

7.1 Confirmed findings

1. **Seven indicators survive held-out validation for breadth.** Of 53 early network indicators screened on DEV, seven are confirmed on held-out field groups (Table 6). They span two families: retained frontier (M0_density_end, D_vol_end, CONTACT_REACH, RETENTION_RATIO) and co-occurrence ego-network (n_comm, NOV_{res}, ego_density). M0_density_end and D_vol_end are partly pre-onset footprint.
2. **The OPEN index.** Six signed, z -scored ego-network components compose into OPEN. The Evaluation 4 descriptive pool of $\text{OPEN}_{\text{home}}$ over 6 non-selection bodies gives partial Spearman +0.069 [+0.038, +0.100], $I^2 = 0$, 6/6 positive. This is a modest association with no practical forecasting gain.
3. **Retained-field relatedness predicts field entry.** On the Experiment 7 independent frame (11,841 concepts, 6,978 entry events), the conditional-logit coefficient is $d_0 = 0.322$ [0.291, 0.355]. The effect is backbone-specific (reverses under min-cp proximity) and the volume-matched contrast is null. Verdict: FRONTIER = PARTIAL.
4. **The consistency–breadth reversal.** Cheng et al.’s ideational consistency predicts volume (+53.5%/SD) but narrower cross-field reach (−0.069). Adding current volume removes 98% of the volume effect. The reversal holds in all five field groups ($I^2 = 0$) and replicates on the 2015–17 cohort.
5. **Breadth is driven by early exploration.** Early contact diversity accounts for the majority of the breadth gap (73% exploration vs 27% retention, variant i_pooled). The frontier-advance component is near zero.
6. **The signal is topical non-redundancy.** Partners from new Leiden communities and mixed-field papers carry the openness–breadth association. The within-concept permutation null absorbs year-to-year churn, though V2 cannot adjudicate temporal turnover at available sample sizes.

7.2 Dead ends and negative results (cumulative)

Table 11: Cumulative dead ends and negative results across all five iterations.

Claim / indicator	Iteration	Deciding evidence
A_h^* naturalisation gap	1	$\Delta\rho = -0.006$; fails all clauses
$D_{\text{ratio}}, F_{\text{res}}$ (dev screen)	1	No candidate survives the rule
G gateway landing on O_{2r}	1	$\Delta\rho = +0.033$; fails
Gateway centrality for retention	2	$\Delta\text{AUC} = -0.00001$ on 27,393 episodes
Gateway weighting for entry	2	Adds nothing beyond plain relatedness
Two-class trajectory typology	2, 4	HMM-vs-DTW ARI = 0.094, 0.222
Ordering (gateway-first)	2	Mixed; negative lead-lag, pre-trend
G-variant O1 gains	2	Label-coverage artefacts
O_5 external recognition	3	No indicator beats B5
Candidate S (co-author)	3	Tested; none confirmed
Rao-Stirling, $G_{\text{btw}}, \log$ offhome	3	Not confirmed on held-out
Experiment 9 (RQ2 trajectories)	3	Failed; never ran
Experiment 11 closure	4, 5	NOT SUPPORTED; opposite sign on heldout
PR2 (localised keep more early)	4	REVERSED
RETENTION_RATIO at R2 on cohort	4	-0.043 [$-0.116, 0.031$]
Sequence ordering	4	No signal beyond mechanical lag
Frame N Cheng reversal	5	Not confirmed
V2 excess churn	5	Cannot adjudicate (too unreliable)
No forecasting gain from network structure	4, 5	B5 Spearman ~ 0.77 ; $+0.002$

7.3 What is still open

- The persistence-filtered RCA rival ($D_{\text{rca_persist_k}}$) differs from Experiment 7's $D_{\text{rca_pers}}$ (max $\rho = 0.877$) and remains untested.
- LIFEENV weakness (PSP $+0.071$ vs $+0.186$ for the other five units pooled): unexplained, not range restriction or coverage.
- The degree-normalised ego density and the Palla size $\times$ turnover interaction were recommended by Research 3 but not run.

- Causal mechanism: the association is correlational. Early ego-network openness could reflect the concept’s intrinsic generality, the diversity of the research community around it, or the breadth of the problems it addresses.
- Domain heterogeneity in field entry ($I^2 = 0.92$).

7.4 Run bookkeeping

The run executed five iterations. Twenty artifacts were commissioned; sixteen completed, four failed or were incomplete (`gen_art_dataset_1`, `gen_art_experiment_2`, `gen_art_experiment_9`, `gen_art_experiment_11` [incomplete]). Every iteration received a reviewer score of 2–3 (blocking). Evidence state moved from *lead* at each hypothesis update, with confidence decreasing at every step. No trajectory log with per-iteration ledger spend was provided for this run.